

Languages

Italian (mother tongue)
English (fluent)
Spanish (beginner)

Core skills

- Robotics & control: **KUKA LBR** Med (**Sunrise.OS**, FRI), **ROS2**, closed-loop PID & force-feedback control.
- Software/Coding: **SolidWorks**, **Java**, **Python**, **C/C++**, MATLAB, **Git**, Arduino, **Office**.
- Regulatory & QA: EU **MDR** 2017/745, ISO **13485**, ISO **14971**, IEC **60601** / **80601**, IEC **62304**.
- Lab & Prototyping: **Ex-vivo** biomechanical testing, **sensor** calibration, PCB **soldering** (incl. SMD), FDM **3D printing**.
- **Licensed Industrial Engineer** (ITA).
- **Scientific writing**.
- **Driving licence**: A, B.

Hobbies & Other activities

Volunteering:

- **Ambulance driver** and first **responder** (2014 - Present);
- **BLSD and safety trainer** for internal and external staff;
- Event **organizer** and **cook** (HACCP certified) in local community events (**Fa'Festa** 2011 - Present).

Personal projects:

- **DIY sous-vide cooker**;
- **Programmable PSU**;
- **Full restoration** of a **Piaggio Si**.

Other interests:

- **Rugby, trekking;**
- **Cooking.**

Lorenzo Maggi

Milan, Italy

Biomedical & Robotics Engineer

Hands-on biomedical & robotics engineer (PhD) with experience in **surgical robotics, knee biomechanics, mechanical design and ex-vivo experimentation.**
Developed a knee kinematic simulator and contributed to the design of **Robin®**, a robotic system for total knee arthroplasty.
Strong **maker** mindset **(mechanics, electronics, 3D printing)** and a focus on **translational** research.

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My Scholar Profile

Experience & Education

R&D Spine Implants Engineer

September 2026 - Present
Clover Orthopedics Srl

- **Mechanical design** and development of **spinal implants**;
- **EU MDR 2017/745 compliant design,** risk analysis, and technical documentation review.

R&D Robotics and Biomedical Engineer

November 2025 - August 2026
Orthokey Italia Srl

- **Co-development** of **Robin®** robotic system for **TKA**;
- **Medical device** design compliant with **ISO 13485, ISO 14971, IEC 60601/80601, and IEC 62304**;
- **KUKA LBR Med** software architecture and control implementation;
- Collaboration with **CNR-IN** on automated murine behavioral analysis via **deep learning**.

PhD in Industrial Engineering (cum Laude)

November 2022 - October 2025
Orthokey Italia Srl - University of Florence

- Thesis: "**Knee-Lax**: an innovative **Robotic Intra-Operative Laxity assessment** module for **Robin** system";
- **Development** of an innovative **knee laxity** assessment robotic tool based on **KUKA LBR Med**;
- **Force-feedback control** implementation for a **rehabilitative hand exoskeleton**;
- **2 years** of Industrial experience at **Orthokey Italia Srl**;
- **Video pitch**: https://youtu.be/5DqW-fMwWBU

Visiting PhD Researcher

September 2024 - March 2025
Université Libre de Bruxelles (Prof. B. Innocenti)

- **Design and validation of KneeKi**, a portable **knee simulator** for kinematic evaluation of **TKA implants**;
- **Closed-loop PID controller** implementation for physiological quadriceps tendon loading;
- Experimental testing campaigns on **ex-vivo cadaveric** knee specimens .

MSc in Biomedical Engineering (110/110 cum Laude)

October 2022 - University of Florence

- Thesis: Design of a compact **hypo & normothermic perfusion device** for the transport and maintenance of explanted **livers.**

BSc in Mechanical Engineering

February 2021 - University of Florence